MSG No. 51.294 $Pm'm'a$ [Type III, orthorhombic]

Table 1: Wyckoff site: `2a`, site symmetry: `.2'/m'.`

No.	position	mapping
1	$[0, 0, 0]$	`[1,3,6,8]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,4,5,7]`

Table 2: Wyckoff site: `2b`, site symmetry: `.2'/m'.`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,3,6,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,4,5,7]`

Table 3: Wyckoff site: `2c`, site symmetry: `.2'/m'.`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,3,6,8]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,4,5,7]`

Table 4: Wyckoff site: `2d`, site symmetry: `.2'/m'.`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,3,6,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,4,5,7]`

Table 5: Wyckoff site: `2e`, site symmetry: `m'm'2`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, z\right]$	`[1,2,7,8]`
2	$\left[\frac{3}{4}, 0, -z\right]$	`[3,4,5,6]`

Table 6: Wyckoff site: `2f`, site symmetry: `m'm'2`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, z\right]$	`[1,2,7,8]`
2	$\left[\frac{3}{4}, \frac{1}{2}, -z\right]$	`[3,4,5,6]`

Table 7: Wyckoff site: `4g`, site symmetry: `.2'.`

No.	position	mapping
1	$[0, y, 0]$	`[1,6]`
2	$\left[\frac{1}{2}, -y, 0\right]$	`[2,5]`

continued ...

Table 7

No.	position	mapping
3	$[0,\ -y,\ 0]$	`[3,8]`
4	$\left[\frac{1}{2},\ y,\ 0\right]$	`[4,7]`

Table 8: Wyckoff site: `4h`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{2}\right]$	`[1,6]`
2	$\left[\frac{1}{2},\ -y,\ \frac{1}{2}\right]$	`[2,5]`
3	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[3,8]`
4	$\left[\frac{1}{2},\ y,\ \frac{1}{2}\right]$	`[4,7]`

Table 9: Wyckoff site: `4i`, site symmetry: `.m'.`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,8]`
2	$\left[\frac{1}{2}-x,\ 0,\ z\right]$	`[2,7]`
3	$[-x,\ 0,\ -z]$	`[3,6]`
4	$\left[x+\frac{1}{2},\ 0,\ -z\right]$	`[4,5]`

Table 10: Wyckoff site: `4j`, site symmetry: `.m'.`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ z\right]$	`[1,8]`
2	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ z\right]$	`[2,7]`
3	$\left[-x,\ \frac{1}{2},\ -z\right]$	`[3,6]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[4,5]`

Table 11: Wyckoff site: `4k`, site symmetry: `m'..`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ z\right]$	`[1,7]`
2	$\left[\frac{1}{4},\ -y,\ z\right]$	`[2,8]`
3	$\left[\frac{3}{4},\ -y,\ -z\right]$	`[3,5]`
4	$\left[\frac{3}{4},\ y,\ -z\right]$	`[4,6]`

Table 12: Wyckoff site: `8l`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[\frac{1}{2}-x,\ -y,\ z\right]$	`[2]`
3	$[-x,\ -y,\ -z]$	`[3]`

continued ...

Table 12

No.	position	mapping
4	$\left[x+\frac{1}{2},\, y,\, -z\right]$	[4]
5	$\left[x+\frac{1}{2},\, -y,\, -z\right]$	[5]
6	$[-x,\, y,\, -z]$	[6]
7	$\left[\frac{1}{2}-x,\, y,\, z\right]$	[7]
8	$[x,\, -y,\, z]$	[8]